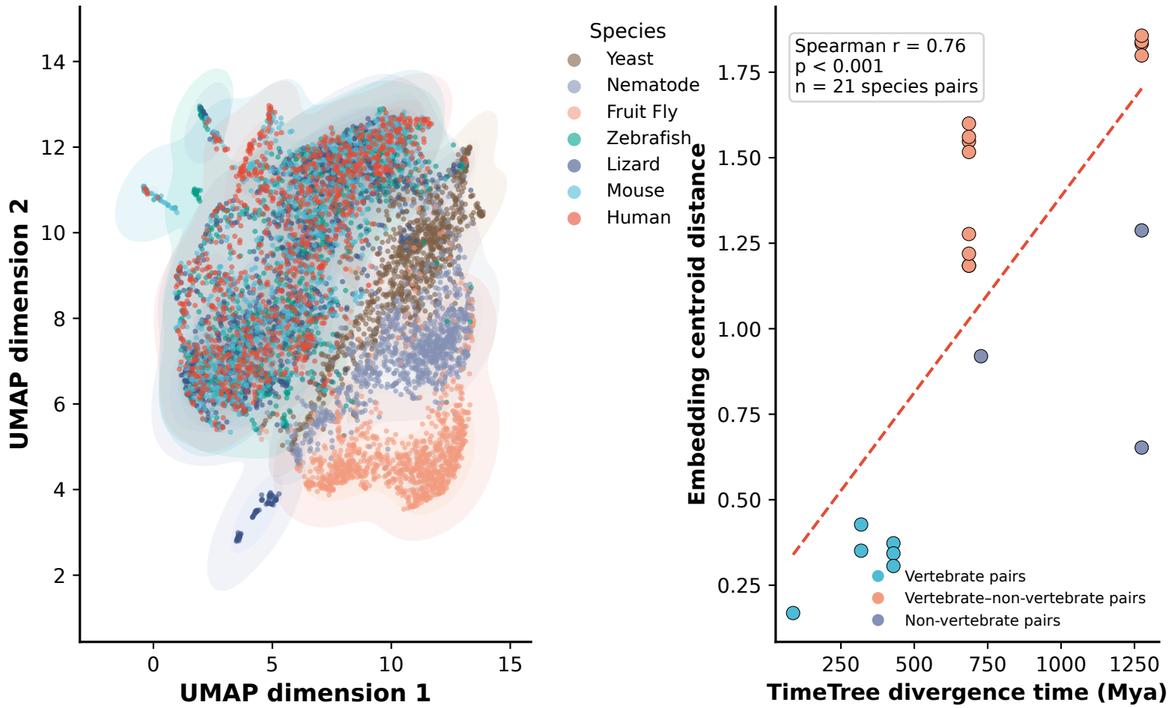
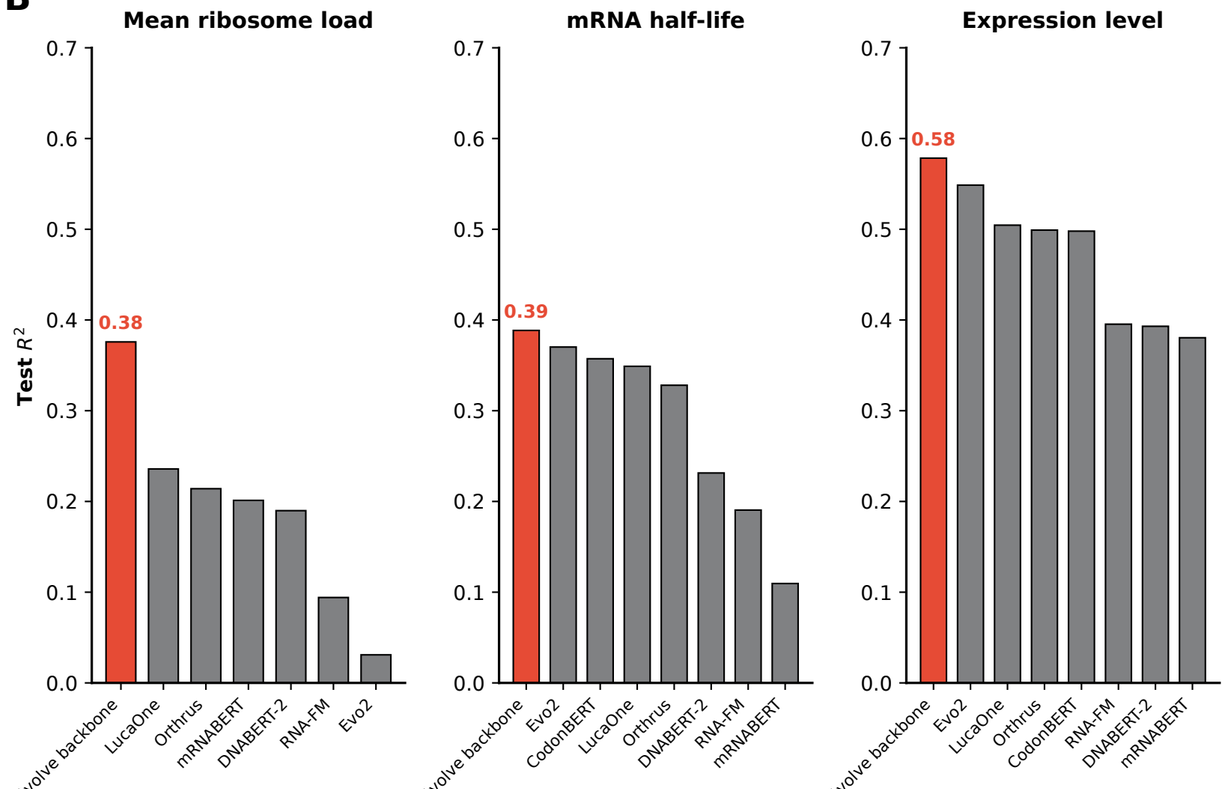
A

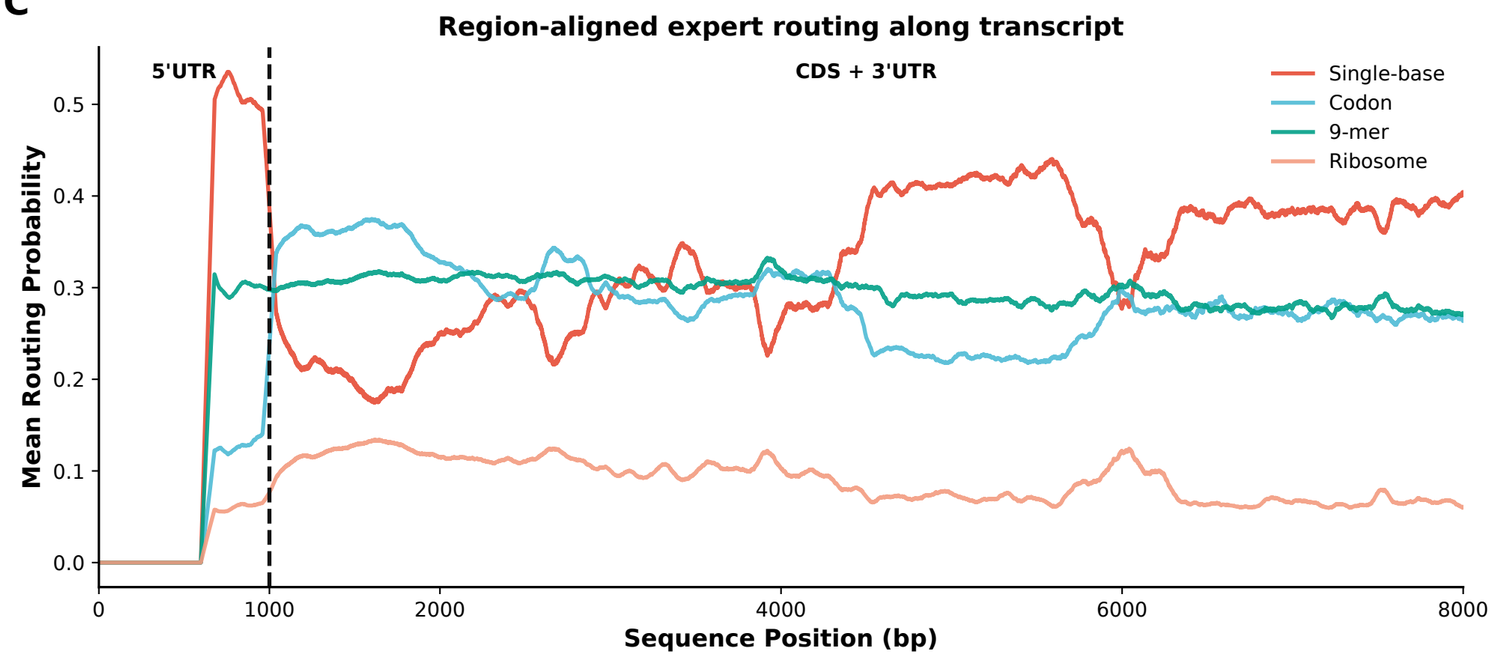

UMAP dimension 2
UMAP dimension 1
Species
Yeast
Nematode
Fruit Fly
Zebrafish
Lizard
Mouse
Human

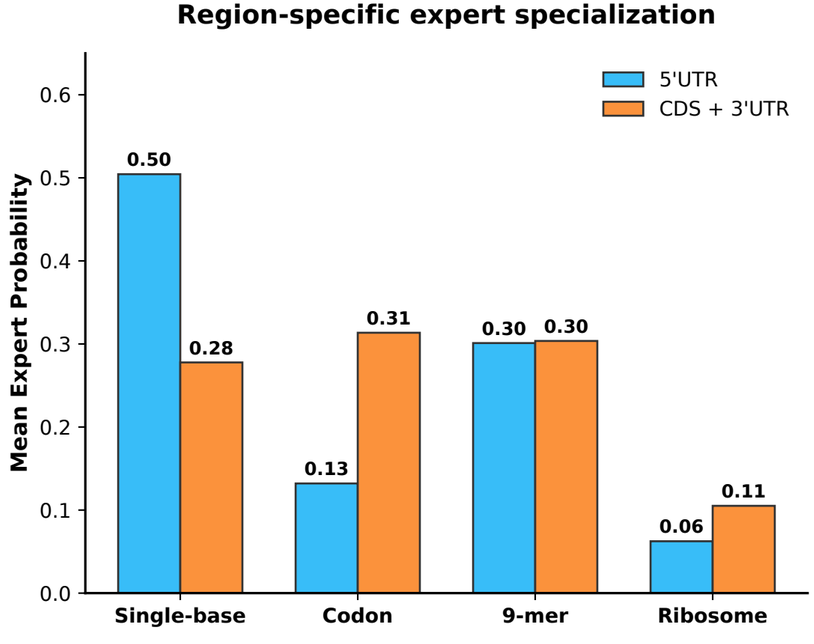
Embedding vs
TimeTree divergence
Spearman r = 0.76
p < 0.001
n = 21 species pairs
Embedding centroid distance
TimeTree divergence time (Mya)
Vertebrate pairs
Vertebrate–non-vertebrate pairs
Non-vertebrate pairs

B


Mean ribosome load
mRNA half-life
Expression level
Test R^2
0.38
0.39
0.58
RNAvolve backbone
LucaOne
Orthrus
mRNABERT
DNABERT-2
RNA-FM
Evo2
CodonBERT

C


Region-aligned expert routing along transcript
5'UTR
CDS + 3'UTR
Single-base
Codon
9-mer
Ribosome
Mean Routing Probability
Sequence Position (bp)


Region-specific expert specialization
5'UTR
CDS + 3'UTR
Mean Expert Probability
0.50
0.28
0.13
0.31
0.30
0.30
0.06
0.11
Single-base
Codon
9-mer
Ribosome